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Linear Algebra

4Week [25 September - 01 October]

Quiz 4

Started on	2023-10-02 07:09
State	Finished
Completed on	2023-10-02 07:59
Time taken	50 mins 47 secs
Marks	4.33/5.00
Grade	1.73 out of 2.00 (87%)

Question 1

Correct

Mark 1.00 out of 1.00

Define a dimension of a space spanned by vectors

$$v_1 = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}.$$

$$v_2 = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}.$$

$$v_3 = \begin{bmatrix} 6 \\ 1 \\ -3 \end{bmatrix}.$$

Select one:

- ☒ 2 ✓
- ☐ 0
- ☐ 3
- ☐ 1
- ☐ 4

Your answer is correct.

The correct answer is: 2

Question 2

Correct

Mark 1.00 out of 1.00

Define a rank of a 5×5 matrix, if it has 3 linearly independent columns.

Select one:

- ☐ 1
- ☐ 2
- ☐ 4
- ☐ 5
- ☒ 3 ✓

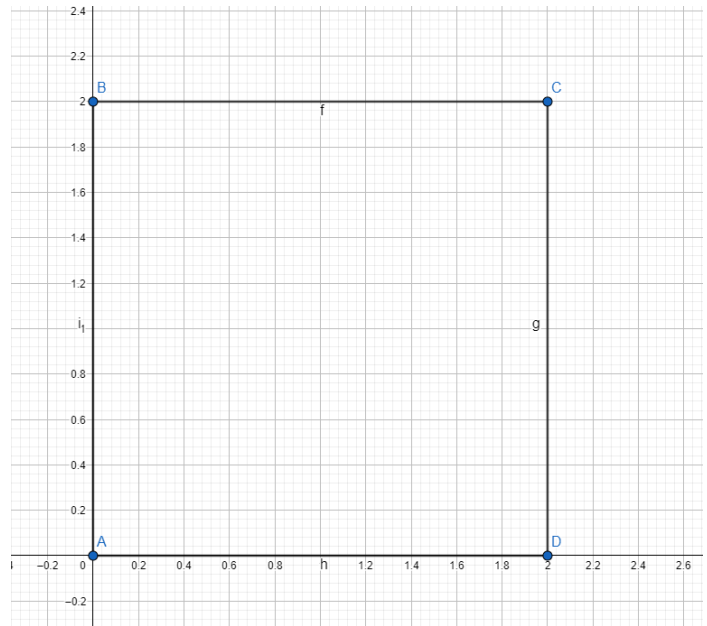
Your answer is correct.

The correct answer is: 3

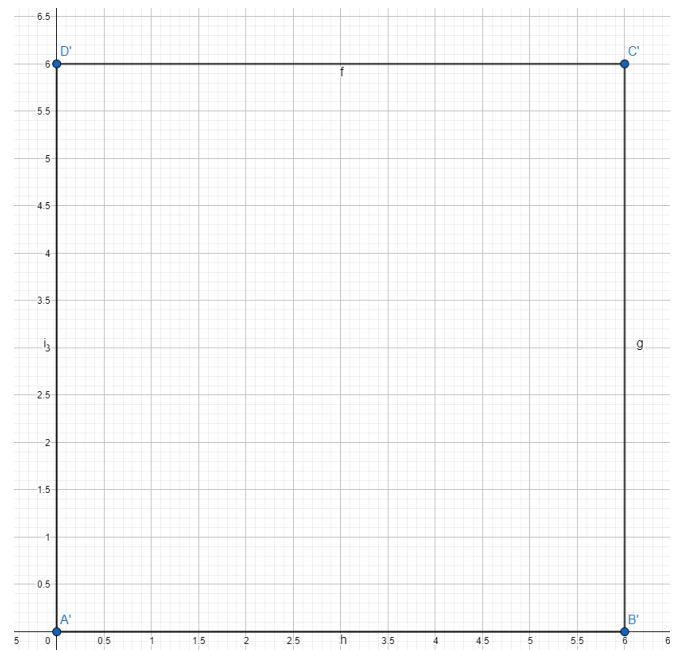
Question 3
Correct
Mark 1.00 out of 1.00

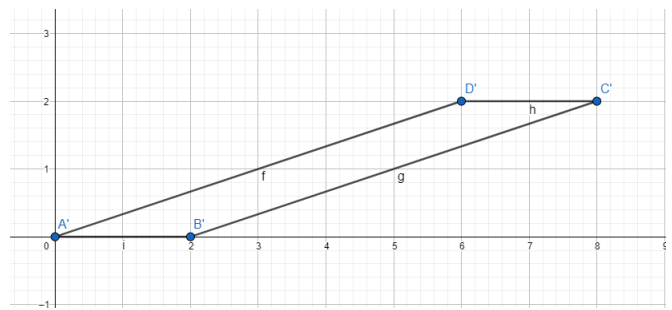
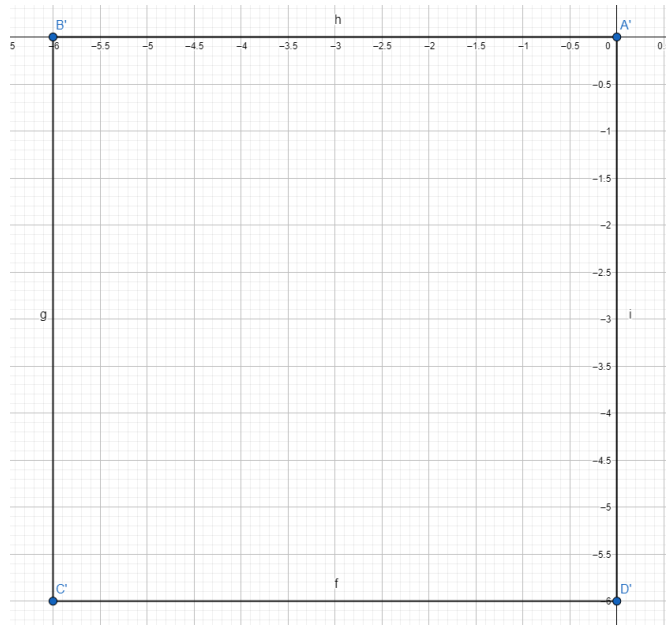
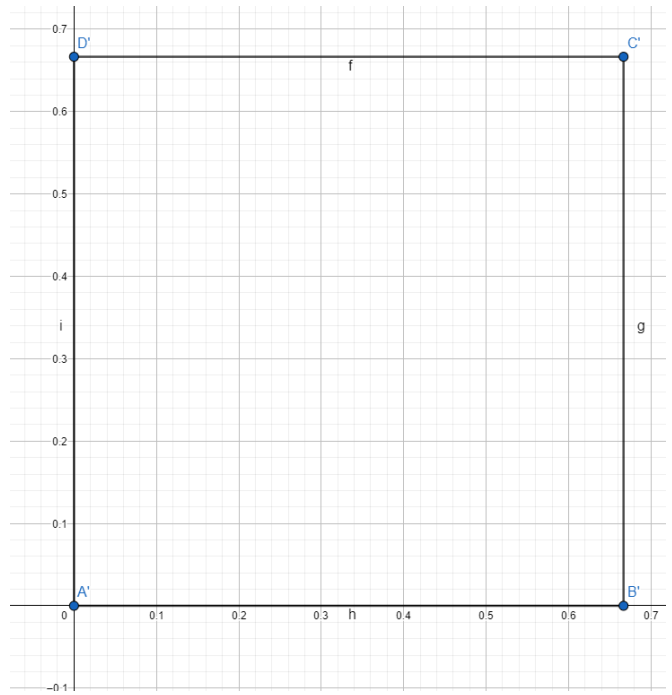
Find the image of given square after shear transformation with

$$k = 3$$



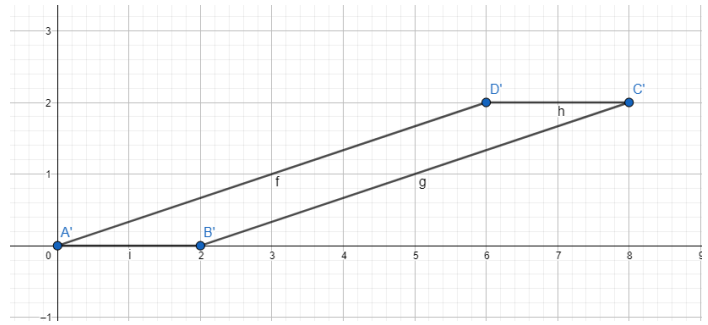
Select one:





Your answer is correct.

The correct answer is:



Question 4
Partially correct
Mark 0.67 out of 1.00

Find true statements.

Select one or more:

- ☐ Matrix with 5 rows and 3 columns may have 4 pivots.
- ☒ If $n \times n$ matrix is invertible, then it has n - pivots. ✓
- ☒ Rank of a matrix is equal to the number of linearly independent columns. ✓
- ☐ If rank of a matrix is equal to 3, then it has at least 3 rows.

Your answer is partially correct.

You have correctly selected 2.

The correct answers are: If $n \times n$ matrix is invertible, then it has

n - pivots. , If rank of a matrix is equal to 3, then it has at least 3

rows., Rank of a matrix is equal to the number of linearly independent columns.

Question 5
Partially correct
Mark 0.67 out
of 1.00

Determine vectors which belong to the

$$\text{span}\left\{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 7 \end{bmatrix}\right\}.$$

Select one or more:

☐ $\begin{bmatrix} 0 \\ -1 \\ 8 \end{bmatrix}$

☒ $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ ✓

☒ $\begin{bmatrix} 2 \\ -1 \\ -21 \end{bmatrix}$ ✓

☐ $\begin{bmatrix} -1 \\ -2 \\ -7 \end{bmatrix}$

Your answer is partially correct.

You have correctly selected 2.

The correct answers are:

$$\begin{bmatrix} -1 \\ -2 \\ -7 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ -21 \end{bmatrix}$$